LAB03: REAL-WORLD DATA LINEAR REGRESSION

PART A: DATA

Dataset: 2024 Population and Gross Regional Domestic Product (GRDP) by Philippine Region

Source: Philippine Statistics Authority (PSA) – OpenSTAT

- **Regional GDP**
  https://openstat.psa.gov.ph/PXWeb/pxweb/en/DB/DB__3S__C3/0502B2BGRD0.px/table/tableViewLayout2/?rxid=e211c484-1754-4c6f-965a-b92d9d936b1d
- **Population per Region**
  https://psa.gov.ph/system/files/phcd/1_2024%20POPCEN%20Press%20Release_Declaration%20of%20Count_Revised_ONS-signed_0.pdf

For the regression:

- **Independent variable (x):** Population, 2024 (persons)
- **Dependent variable (y):** GRDP, 2024 (thousand Philippine pesos)
- **Number of observations:** 18 regions

Description: The dataset compares the population and economic output of the different regions of the Philippines. Linear regression will be used to determine whether population size is associated with regional economic output.

No	Philippine Region	Population, 2024 (persons)	GRDP, 2024 (thousand PHP)
1	National Capital Region (NCR)	14,001,751	8,214,308,357
2	Cordillera Administrative Region (CAR)	1,808,985	445,523,963
3	Ilocos Region (Region I)	5,342,453	877,318,738

4	Cagayan Valley (Region II)	3,777,608	564,165,184
5	Central Luzon (Region III)	12,989,074	2,892,168,121
6	CALABARZON (Region IV-A)	16,933,234	3,723,568,508
7	MIMAROPA Region	3,245,446	509,501,135
8	Bicol Region (Region V)	6,064,426	769,941,504
9	Western Visayas (Region VI)	4,861,911	797,040,847
10	Negros Island Region (NIR)	4,904,944	806,010,142
11	Central Visayas (Region VII)	6,640,875	1,491,366,498
12	Eastern Visayas (Region VIII)	4,625,929	621,904,785
13	Zamboanga Peninsula (Region IX)	3,943,837	582,571,225
14	Northern Mindanao (Region X)	5,178,326	1,273,436,870
15	Davao Region (Region XI)	5,389,422	1,376,650,876
16	SOCCSKSARGEN (Region XII)	4,462,776	675,617,883
17	Caraga (Region XIII)	2,865,196	439,146,876

18	Bangsamoro Autonomous Region in Muslim Mindanao (BARMM)	5,691,583	386,127,983

PART B: PYTHON SCRIPT

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# Numerical Methods - Laboratory Exercise 03
# Real-World Data Linear Regression
# Philippine Regional Population vs. GRDP, 2024

# Import NumPy for numerical calculations
import numpy as np

# This is the dataset
# Independent variable: Population of each region
# Unit: persons

x = np.array([
    14001751,
    1808985,
    5342453,
    3777608,
    12989074,
    16933234,
    3245446,
    6064426,
    4861911,
    4904944,
    6640875,
    4625929,
    3943837,
    5178326,
    5389422,
    4462776,
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    2865196,
    5691583
], dtype=float)

# Dependent variable: GRDP of each region
# Unit: thousand Philippine pesos
y = np.array([
    8214308357,
    445523963,
    877318738,
    564165184,
    2892168121,
    3723568508,
    509501135,
    769941504,
    797040847,
    806010142,
    1491366498,
    621904785,
    582571225,
    1273436870,
    1376650876,
    675617883,
    439146876,
    386127983
], dtype=float)

# Number of observations
n = len(x)

# Calculating the required summations

sum_x = np.sum(x)
sum_y = np.sum(y)
sum_xy = np.sum(x * y)
sum_x2 = np.sum(x ** 2)
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# Calculating the slope (a1)

a1 = (n * sum_xy - sum_x * sum_y) / \
     (n * sum_x2 - sum_x ** 2)

# Calculate the y-intercept (a0)

a0 = (sum_y - a1 * sum_x) / n

# The results

print("Number of observations:", n)
print("Sum of x:", sum_x)
print("Sum of y:", sum_y)
print("Sum of xy:", sum_xy)
print("Sum of x^2:", sum_x2)

print("\nLeast-Squares Regression Results")
print("a0 (intercept) =", a0)
print("a1 (slope)     =", a1)

# The final regression equation
print("\nRegression equation:")
print(f"y = {a0:.4f} + ({a1:.4f})x")
```

PART C: REGRESSION RESULT

Statistics	Result
Number of observations, nn	18
Intercept, a_0	− 897,518,575.2573
Slope, a_1	377.9166534
Sr / SSE	$2.0969230595 \times 10^{19}$
r^2	0.6592455433
Standard error, $S_{y/x}$	1,144,804,311.7574

Regression Equation

The fitted model is:

$y = a_0 + a_1 x$

With:

$a_0 = -897{,}518{,}575.2573$

$a_1 = 377.9166534$

Therefore, the fitted regression equation is:

$y = -897{,}518{,}575.2573 + 377.9166534x$

where:

x = regional population (persons)

y = GRDP (thousand PHP)

For Sr (Sum of Squared Residuals) or SSE (Sum of Squared Errors):

$S_r = \sum(y - \hat{y})^2$

$S_r = 2.0969230595 \times 10^{19}$

where:

y = actual value

ŷ = predicted value from the regression line

$y - \hat{y}$ = residual error

For r^2:

The coefficient of determination is calculated using:

$r^2 = 1 - Sr/St$

where:

$St = \sum(y - \bar{y})^2$ (is the total sum of squares)

For the dataset:

$r^2 = 0.6592$ or 65.92%

For Standard Error ($S_{y/x}$):

The standard error is calculated using:

$$S_{y/x} = \sqrt{\frac{Sr}{n-2}}$$

We have:

$S_r = 2.0969230595 \times 10^{19}$

and:

n=18

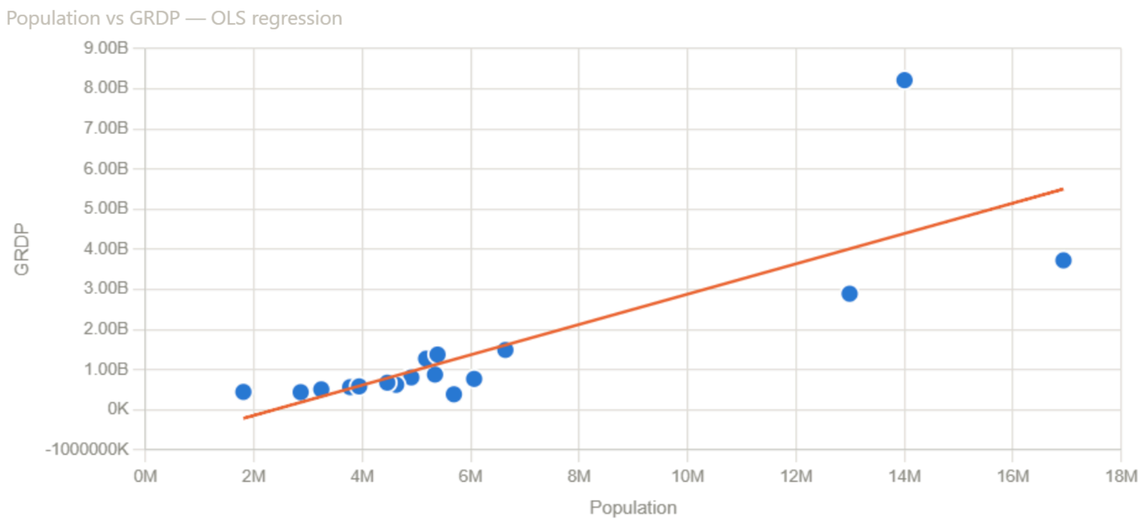
Therefore:

$$S_{y/x} = \sqrt{\frac{2.0969230595x10^{19}}{18-2}}$$

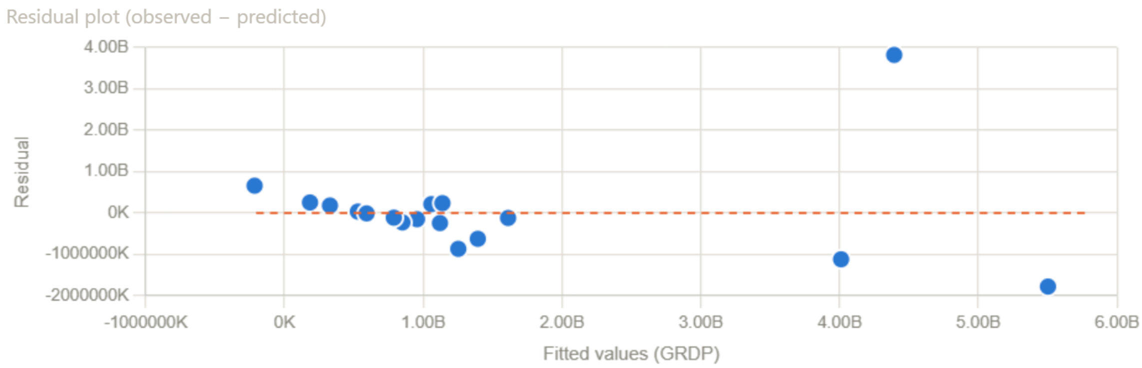
So:

$S_{y/x}$ = 1,144,804,311.76

FITTED REGRESSION LINE



RESIDUAL PLOT



R^2 = 0.6592 | slope = 377.9167 | intercept = -897519K

Residuals from the 2024 Philippine regional Population vs. GRDP least-squares regression. The residuals show the difference between the actual GRDP and the predicted GRDP. Some residuals are positive while others are negative. This means that the regression line sometimes predicts a value higher than the actual GRDP and sometimes predicts a lower value. The large residuals of some regions show that population alone cannot completely predict GRDP.

INTERPRETATION

The slope of the regression line is:

$a_1 = 377.9167$

This means that regions with a larger population generally have a higher GRDP. Based on the model, an increase in population is associated with an increase in GRDP.

The intercept is:

$a_0 = -897{,}518{,}575.2573$

This is the estimated GRDP when the population is zero. Since a region cannot realistically have zero population, the intercept does not have much practical meaning.

The value of r^2 is:

$r^2 = 0.6592$ or 65.92%.

This means that the population explains about 65.92% of the changes in GRDP in the dataset. Therefore, the regression line shows a fairly good relationship between population and GRDP, although other factors also affect GRDP.

The standard error is:

$S_{y/x} = 1{,}144{,}804{,}311.76$ thousand PHP

This shows that the actual GRDP values can differ considerably from the values predicted by the regression line.

PREDICTION

For the prediction, let's use a population of:

$x = 7{,}000{,}000$ persons

This value was not included in the original dataset.

Using the regression equation:

$y = -897{,}518{,}575.2573 + 377.9166534x$

Substitute x=7,000,000:

$y = -897{,}518{,}575.2573 + 377.9166534(7{,}000{,}000)$

$y = 1{,}747{,}897{,}998.54$ thousand PHP

Therefore:

$y \approx$ ₱1.748 trillion

If a Philippine region has a population of 7 million people, the regression model predicts that its GRDP would be approximately ₱1.748 trillion.